CS4433 – Moderated Learning Management System

Installation Guide

Version: 1.0
Application: Moderated Learning Management System
Programming Language: C++17
Database: SQLite
Build System: CMake
Platform: Windows 10 / Windows 11

1. System Overview

The Moderated LMS is a C++ console-based application designed to manage academic activities within an educational institution.

The system supports four categories of users:

- Students
- Lecturers
- Heads of Department (HODs)
- Administrators

It provides functionality for user authentication, course enrolment, assignment management, assignment submissions, grading, moderation and result release.

The application uses an SQLite database to store system information.

2. System Requirements

Before installing the application, ensure that the computer meets the following requirements.

2.1 Hardware Requirements

- A Windows-compatible desktop or laptop.
- Sufficient storage space for the application and database.
- Sufficient memory to run the application and operating system.
- Keyboard and monitor.

2.2 Software Requirements

Software	Requirement	Purpose
Operating System	Windows 10 or Windows 11	Application environment
C++ Compiler	GCC with C++17 support	Compiling the application
CMake	Version 3.20 or later	Project configuration and build
MSYS2	UCRT64 environment	C++ development toolchain
SQLite3	Required development library	Database operations
libsodium	Required development library	Password hashing and verification
Git	Optional	Obtaining the source code

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3. Obtaining the Application

The project source code is hosted on GitHub.

Repository:

https://github.com/Kevin-droid408/CS4433-Moderated-Learning-Management-System

Open PowerShell and execute:

git clone https://github.com/Kevin-droid408/CS4433-Moderated-Learning-Management-System.git

Navigate to the project directory:

cd CS4433-Moderated-Learning-Management-System

For the v1.0 release, the technician should check out the required release tag:

git checkout v1.0

This ensures that the version being installed corresponds to the approved release.

4. Installing Dependencies

The application requires a C++17-compatible compiler, CMake, SQLite3 and libsodium.

4.1 Install MSYS2

Download and install MSYS2 from its official website:

https://www.msys2.org/

After installation, open the MSYS2 UCRT64 terminal.

Update the package database:

```
pacman -Syu
```

If the terminal requests a restart, close it, reopen the UCRT64 terminal and complete the update.

4.2 Install the Required Packages

Execute:

```
pacman -S --needed \
  mingw-w64-ucrt-x86_64-gcc \
  mingw-w64-ucrt-x86_64-cmake \
  mingw-w64-ucrt-x86_64-make \
  mingw-w64-ucrt-x86_64-sqlite3 \
  mingw-w64-ucrt-x86_64-libsodium \
  mingw-w64-ucrt-x86_64-pkgconf
```

These packages provide the compiler, build tools and required libraries.

4.3 Verify the Installation

Check the compiler:

```
g++ --version
```

Check CMake:

```
cmake --version
```

Check Make:

```
mingw32-make --version
```

The commands should display the installed versions.

5. Configuring the Application

Open the MSYS2 UCRT64 terminal in the project directory.

Configure the project using CMake:

cmake -S . -B build -G "MinGW Makefiles"

CMake will:

1. Detect the installed C++ compiler.
2. Configure the C++17 build environment.
3. Locate SQLite3.
4. Locate libsodium where required by the project.
5. Generate the build files.

If configuration completes successfully, the project is ready to compile.

6. Building the Application

Execute:

cmake --build build

CMake will compile the application and the executable targets defined in the project's CMake configuration.

Check the build output for errors.

The main application executable is expected to be:

build/

ModeratedLMS.exe

The project also provides separate setup utilities, including:

- CreateAdministrator.exe
- CreateHOD.exe
- CreateLecturer.exe
- SetupDepartments.exe

The exact executable targets depend on the project's CMake configuration.

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7. Database Configuration

7.1 Database Overview

The LMS uses SQLite as its relational database management system.

The database stores information relating to:

- Users
- Departments
- Courses
- Course enrolments
- Assignments
- Assignment extensions
- Student submissions
- Grades and feedback
- Moderation records and audit information

The database schema is located at:

database/schema.sql

The application initialises the database using this schema when required.

7.2 Database Location

The database file is:

database/lms.db

Ensure that the application has permission to access and modify the database directory.

7.3 Database Protection

The database contains academic and user information.

The technician should:

- Restrict access to the database directory.
- Prevent unauthorised modification of database records.
- Back up the database regularly.
- Store backups securely.
- Avoid deleting the database during upgrades.

8. Creating the Initial Administrator Account

The system requires an Administrator account for managing users, departments and courses.

Run the administrator setup utility from the project root:

.\build\CreateAdministrator.exe

Follow the displayed instructions.

The utility prompts for the required account information and password.

After successful creation, the Administrator can use the account to access the LMS.

The administrator credentials should be kept confidential.

And an existing Administrator Account:

Username: admin

Password: 1470@kevin

9. Launching the Application

To launch the LMS, execute:

.\build\ModeratedLMS.exe

The application displays its login interface.

Enter the registered account credentials.

After successful authentication, the system displays the menu corresponding to the user's role.

10. Troubleshooting

Problem	**Possible cause**	**Recommended solution**
Application fails to launch	Missing runtime dependency	Ensure required runtime libraries are installed and accessible
CMake cannot find the compiler	Incorrect environment	Open the MSYS2 UCRT64 terminal
SQLite3 cannot be found	Missing development package	Install the SQLite3 development package
libsodium cannot be found	Missing dependency	Install the libsodium development package

Build fails	Configuration or source-code error	Review the CMake and compiler output
Database cannot be opened	Incorrect path or insufficient permissions	Verify the database location and directory permissions
Login is unsuccessful	Incorrect credentials or account not registered	Verify the account details
Administrator utility cannot be found	Target was not built	Rebuild the project and check the CMake targets

11. Support and Project Information

Project: CS4433 Moderated Learning Management System

Release: v1.0

Repository:

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